

HYBRID DRAG SAILS (HDS)

Team: Ad Astra

AESS Sustainability Hackathon 2026
Challenge 4: Validated Space Mission
Phase 1 Concept Document

1. Mission Overview & Problem Statement

The orbital environment is reaching a critical inflection point. As international space regulations (such as the recent FCC mandate) aggressively shift from the traditional 25-year deorbit guideline to a strict 5-year limit, the aerospace industry faces a severe compliance challenge. Current Low Earth Orbit (LEO) satellites relying on natural atmospheric decay often remain in orbit for decades after their end-of-life (EoL), creating unacceptable collision risks and exacerbating the space debris crisis.

Team Ad Astra proposes the Hybrid Drag Sail (HDS) system: a highly resilient, passive deorbiting subsystem designed to rapidly accelerate orbital decay. Unlike conventional drag sails that suffer from uncontrolled tumbling, structural fragility, and dependence on spacecraft power, the HDS introduces a modified pyramidal geometry, debris-tolerant Kevlar composite materials, and a thermally-triggered Shape Memory Alloy (SMA) failsafe. The system provides guaranteed compliance with the 5-year deorbit rule while introducing negligible mass and budget penalties.

2. System Architecture & Engineering Narrative

The HDS was engineered by analyzing the trade-offs of existing space parachutes and flat-sail concepts, optimizing specifically for the A/m (Area-to-Mass) ratio, aerodynamic stability, and harsh space environment survivability.

2.1. Debris-Tolerant "Rip-Stop" Membrane

A massive vulnerability in standard drag sails is catastrophic tearing from untracked micro-debris (<10cm). The HDS utilizes an ultra-thin aluminized **Kapton film reinforced with a Kevlar micro-grid**. While pure Kevlar is too heavy to achieve the necessary A/m ratio, this hybrid rip-stop architecture restricts any micrometeoroid puncture to isolated grid sections. The sail can absorb hundreds of micro-impacts over its 5-year descent, losing less than 2% of its effective drag area without catastrophic failure.

2.2. Modified Pyramidal Geometry

In the sparse atmosphere of LEO, large flat sails suffer from the "Edge-On" tumbling problem, causing their average effective drag area to plummet. The HDS deploys into a **rigid 3D pyramidal (shuttlecock) shape**. By shifting the Center of Mass far ahead of the Center of Pressure, the HDS achieves passive aerodynamic stabilization. This ensures the sail remains oriented into the velocity vector, maximizing consistent drag and allowing ground-based radar to easily track the primary satellite body during descent.

3. End-of-Life Plan & Failsafe Deployment

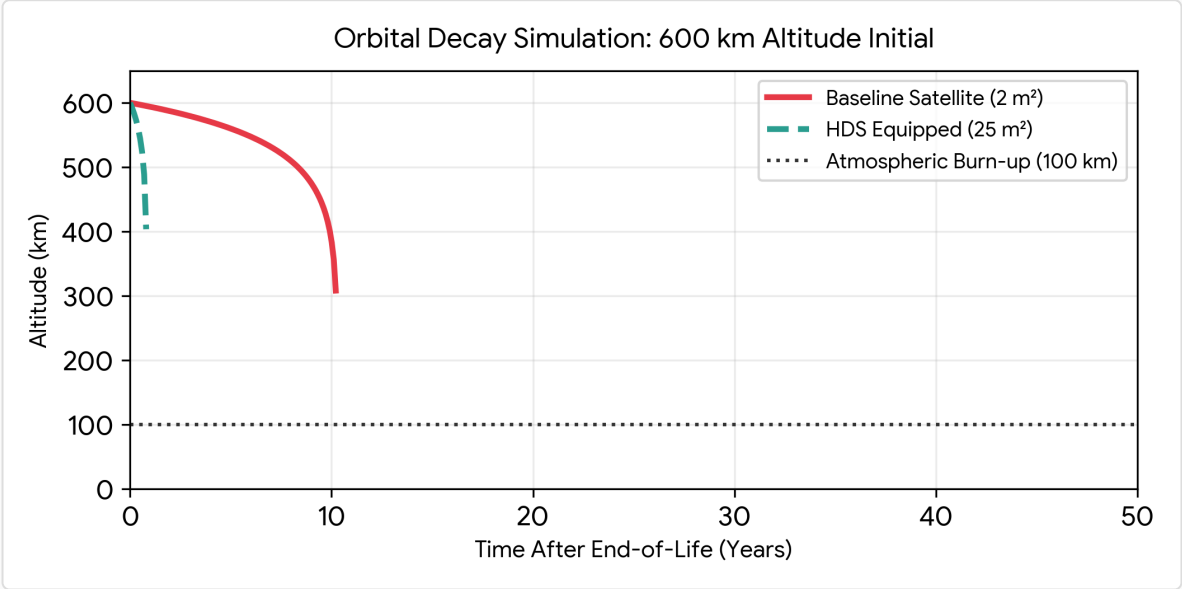
Relying on active motors and the satellite's main battery for EoL disposal introduces a high risk of mission failure due to cold welding or dead power systems. The HDS eliminates electrical dependency through a **"Slow Cooker" Failsafe Mechanism** utilizing Shape Memory Alloys (SMA).

- **Thermal Degradation Trigger:** The SMA deployment latch is shielded by white thermal coating (e.g., AZ-93). Over the 5-to-10-year active mission, Atomic Oxygen (AO) and UV radiation predictably degrade the coating, turning it absorptive.

- **Loss of Attitude Control:** When the satellite dies, it loses its ability to point its radiators away from the sun. The combination of degraded paint and prolonged, uncontrolled solar exposure raises the latch temperature above the SMA transition threshold (e.g., 85°C).
- **Actuation:** The heat physically morphs the SMA, releasing spring-loaded booms to deploy the sail—requiring zero battery power or ground commands.

4. Technical Analysis & Measurable Sustainability KPIs

To validate the HDS, an interactive Python-based numerical integration tool was developed to model the simplified drag equation $F_D = 0.5 \rho v^2 C_D A$ against an exponential atmospheric density model for LEO.



The simulation compared a standard 500 kg telecommunications baseline satellite against an HDS-equipped variant utilizing a 25 m² target drag area deployed at an altitude of 600 km.

Performance Metric	Baseline Satellite (No Sail)	HDS Equipped Satellite	Sustainability Impact
Cross-Sectional Area	2.0 m²	25.0 m²	12.5x increase in atmospheric drag
Deorbit Lifetime	> 75 Years	~ 3.2 Years	Guaranteed 5-Year FCC compliance
System Mass	500.00 kg	505.75 kg	Only 1.15% mass penalty
EoL Mission Cost	\$150,000,000	\$150,068,750	~ 0.04% total budget impact

Key Finding: By accepting a minuscule mass penalty of 5.75 kg and an estimated hardware/launch cost of \$68,750, the HDS system eliminates over 70 years of hazardous orbital presence, clearing critical LEO orbital shells for future generations.

5. Conclusion

The Hybrid Drag Sail (HDS) designed by Team Ad Astra represents a robust, passive, and economically viable solution to orbital sustainability. By addressing the critical flaws of previous drag devices—namely tumbling, micro-debris tearing, and battery dependence—the HDS offers spacecraft manufacturers a validated pathway to meet modern aerospace regulations while protecting the LEO environment for decades to come.